

The LOGISTIC Procedure

Model Information	
Data Set	WORK.MODEL_DATA
Response Variable	Default_Risk
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	800
Number of Observations Used	800

Response Profile		
Ordered Value	Default_Risk	Total Frequency
1	1	564
2	0	236

Probability modeled is Default_Risk='1'.

Class Level Information					
Class	Value	Design Variables			
Checking_Status	1	1	0	0	
	2	0	1	0	
	3	0	0	1	
	4	0	0	0	
Credit_History	0	1	0	0	0
	1	0	1	0	0
	2	0	0	1	0
	3	0	0	0	1
Savings_Amount	4	0	0	0	0
	1	1	0	0	0
	2	0	1	0	0
	3	0	0	1	0
	4	0	0	0	1
	5	0	0	0	0

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	972.509	802.543
SC	977.194	872.812
-2 Log L	970.509	772.543

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	197.9657	14	<.0001
Score	180.6375	14	<.0001
Wald	138.7912	14	<.0001

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Duration_Months	1	9.2568	0.0023
Credit_Amount	1	1.7846	0.1816
Age_Years	1	2.3716	0.1236
Checking_Status	3	60.6237	<.0001
Credit_History	4	28.7414	<.0001
Savings_Amount	4	16.5972	0.0023

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	3.4425	0.4886	49.6426	<.0001
Duration_Months		1	-0.0279	0.00917	9.2568	0.0023
Credit_Amount		1	-0.00005	0.000039	1.7846	0.1816
Age_Years		1	0.0126	0.00816	2.3716	0.1236
Checking_Status	1	1	-1.8213	0.2457	54.9263	<.0001
Checking_Status	2	1	-1.3659	0.2450	31.0853	<.0001
Checking_Status	3	1	-0.4944	0.4353	1.2897	0.2561
Credit_History	0	1	-1.5294	0.4473	11.6882	0.0006
Credit_History	1	1	-1.9164	0.4056	22.3252	<.0001
Credit_History	2	1	-0.5996	0.2282	6.9018	0.0086
Credit_History	3	1	-0.5519	0.3541	2.4288	0.1191
Savings_Amount	1	1	-0.8748	0.2672	10.7228	0.0011
Savings_Amount	2	1	-0.4762	0.3713	1.6446	0.1997
Savings_Amount	3	1	-0.2630	0.4876	0.2908	0.5897
Savings_Amount	4	1	0.5506	0.6130	0.8068	0.3691

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Duration_Months	0.972	0.955	0.990
Credit_Amount	1.000	1.000	1.000
Age_Years	1.013	0.997	1.029
Checking_Status 1 vs 4	0.162	0.100	0.262
Checking_Status 2 vs 4	0.255	0.158	0.412
Checking_Status 3 vs 4	0.610	0.260	1.432
Credit_History 0 vs 4	0.217	0.090	0.521
Credit_History 1 vs 4	0.147	0.066	0.326
Credit_History 2 vs 4	0.549	0.351	0.859
Credit_History 3 vs 4	0.576	0.288	1.153
Savings_Amount 1 vs 5	0.417	0.247	0.704
Savings_Amount 2 vs 5	0.621	0.300	1.286
Savings_Amount 3 vs 5	0.769	0.296	1.999
Savings_Amount 4 vs 5	1.734	0.522	5.766

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	79.7	Somers' D	0.593
Percent Discordant	20.3	Gamma	0.593
Percent Tied	0.0	Tau-a	0.247
Pairs	133104	c	0.797



